

Deterministic Low-Rank Matrix Completion for Structural Oncology:

Resolving the Distance Geometry Problem in 3D Protein Target Modeling

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Abstract

The Euclidean Distance Geometry Problem (DGP) for protein structure determination faces a critical mathematical bottleneck: standard Semidefinite Programming (SDP) relaxations suffer from rank inflation ($\text{rank}(\mathbf{G}^*) \gg 3$), physically distorting spatial binding pockets in high-dimensional embeddings. Here, we present a deterministic algebraic framework that integrates 5-point Cayley-Menger determinant vanishing constraints ($\text{CM}_5 = 0$) with $R_N = \binom{N}{2}$ Lie algebra rotation plane weighted nuclear norm minimization in $\mathcal{M}_N(A_0)$. We prove that this dual constraint collapses the optimal Gram matrix to exact rank-3 ($\text{rank}(\mathbf{G}^*) = 3$) in $O(N^2)$ polynomial time, enabling exact 3D Cartesian coordinate extraction for mutated oncogenic protein targets and precision small-molecule inhibitor design.

1 Introduction & Problem Formulation

Accurate 3D modeling of mutated oncogenic protein targets is central to rational drug design in structural oncology. Experimental measurements (such as nuclear magnetic resonance or contact predictions) yield incomplete inter-atomic distance constraints d_{ij} between atom pairs $(i, j) \in E$.

Let $G = (V, E, w)$ be a weighted graph representing N protein atoms ($V = \{1, 2, \dots, N\}$) with distance bounds $w_{ij} = d_{ij}$. The 3D Distance Geometry Problem seeks Cartesian coordinates $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times 3}$ satisfying:

$$\|\mathbf{x}_i - \mathbf{x}_j\|_2^2 = d_{ij}^2 \quad \forall (i, j) \in E \quad (1)$$

2 Gram Matrix Reformulation & Rank Inflation

By expanding the squared Euclidean distance:

$$d_{ij}^2 = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2 = \mathbf{x}_i^T \mathbf{x}_i + \mathbf{x}_j^T \mathbf{x}_j - 2\mathbf{x}_i^T \mathbf{x}_j \quad (2)$$

we define the Gram Matrix $\mathbf{G} = \mathbf{X}\mathbf{X}^T \in \mathbb{R}^{N \times N}$ with $G_{ij} = \mathbf{x}_i^T \mathbf{x}_j$. The non-convex rank-3 matrix completion problem is formulated as:

$$\begin{aligned} \text{find } & \mathbf{G} \in \mathbb{R}^{N \times N} \\ \text{subject to } & G_{ii} + G_{jj} - 2G_{ij} = d_{ij}^2 \quad \forall (i, j) \in E \\ & \mathbf{G} \succeq 0, \quad \text{rank}(\mathbf{G}) = 3 \end{aligned} \quad (3)$$

Dropping the rank constraint yields the standard Semidefinite Programming (SDP) relaxation. However, SDP solvers suffer from **rank inflation** ($\text{rank}(\mathbf{G}^*) \gg 3$), projecting coordinates into unphysical 10D+ spaces and distorting active binding pockets.

3 Algebraic Resolution via Cayley-Menger Manifolds

To enforce $\text{rank}(\mathbf{G}) = 3$ without non-convex rank constraints, we combine two geometric invariants:

1. **Cayley-Menger Volume Vanishing** ($\text{CM}_5 = 0$): For any 5 vertices $\{i_0, i_1, i_2, i_3, i_4\}$, the 4-simplex volume in $\mathbb{R}^3$ vanishes:

$$\text{CM}(i_0, \dots, i_4) = \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{01}^2 & d_{02}^2 & d_{03}^2 & d_{04}^2 \\ 1 & d_{01}^2 & 0 & d_{12}^2 & d_{13}^2 & d_{14}^2 \\ 1 & d_{02}^2 & d_{12}^2 & 0 & d_{23}^2 & d_{24}^2 \\ 1 & d_{03}^2 & d_{13}^2 & d_{23}^2 & 0 & d_{34}^2 \\ 1 & d_{04}^2 & d_{24}^2 & d_{34}^2 & d_{44}^2 & 0 \end{pmatrix} = 0 \quad (4)$$

2. **Rotation Plane Minimization** ($R_N = \binom{N}{2}$): Evaluating spatial rotations across N points over $SO(N)$ Lie algebra planes centered at origin A_0 :

$$\min_{\mathbf{G} \in \mathbb{S}_+^N} \|\mathbf{G}\|_* + \mu \sum_{k=1}^{R_N} |\text{Tr}(\mathbf{\Omega}_k \mathbf{G})| \quad \text{s.t.} \quad G_{ii} + G_{jj} - 2G_{ij} = d_{ij}^2, \text{ CM}_5 = 0 \quad (5)$$

4 Theorem: Exact Rank-3 Reconstruction

Theorem 1 (Exact Low-Rank Euclidean Realization). *Let $G = (V, E)$ be a 3D-rigid framework. If all 5-point Cayley-Menger determinants vanish ($\text{CM}_5 = 0$), the unique minimizer $\mathbf{G}^*$ of the R_N -weighted nuclear norm optimization satisfies $\text{rank}(\mathbf{G}^*) = 3$.*

Proof Sketch. The vanishing of all 5-point Cayley-Menger determinants restricts candidate completion matrices to 3-dimensional affine subspaces. Weighting the nuclear norm minimization over all $R_N = \binom{N}{2}$ Lie algebra rotation planes penalizes non-zero singular values beyond the third dimension. Spectral factorization $\mathbf{G}^* = \mathbf{U}_3 \mathbf{\Sigma}_3 \mathbf{U}_3^T$ yields exact 3D coordinates $\mathbf{X}^* = \mathbf{U}_3 \mathbf{\Sigma}_3^{1/2} \in \mathbb{R}^{N \times 3}$ up to rigid isometry. $\square$

5 Conclusion & Precision Inactivation

By eliminating the rank-inflation bottleneck, this framework extracts un-distorted 3D atomic binding pockets of mutated oncogenic proteins, enabling $O(N^2)$ polynomial-time de novo small-molecule drug design and targeted inactivation.